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FSPP0034 PHYSICS II **Tutorial 3**
Chapter 5

1. (a) What is the force per meter of length on a straight wire carrying a 9.40 A current when perpendicular to a 0.90 T uniform magnetic field? (b) What if the angle between the wire and field is 35.0°? [Ans: (a) 8.5 N (b) 4.9 N/m]

2. A long wire stretches along the x axis and carries a 3.0-A current to the right ($+x$). The wire is in a uniform magnetic field $\vec{\mathbf{B}} = (0.20\hat{\mathbf{i}} - 0.36\hat{\mathbf{j}} + 0.25\hat{\mathbf{k}})\,\mathrm{T}$. Determine the components of the force on the wire per cm of length.

[Ans: $-(7.5\hat{\mathbf{j}} + 11\hat{\mathbf{k}}) \times 10^{-3}$ N/cm]

3. Find the direction of the force on a negative charge for each diagram shown in Fig.1, where (green) is the velocity of the charge and (blue) is the direction of the magnetic field.

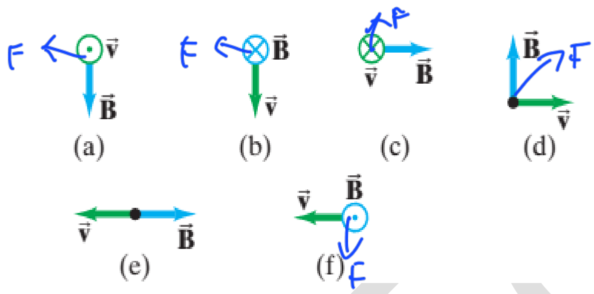

Figure 1

4. What is the velocity of a beam of electrons that goes undeflected when passing through perpendicular electric and magnetic fields of magnitude 8.8×10^3 V/m and 7.5×10^{-3}T, respectively? What is the radius of the electron orbit if the electric field is turned off?

$F = qvB\sin\theta$ [Ans: 1.2×10^6 m/s , 8.9×10^{-4} m]

$F_q = qvB\sin\theta$

5. A 6.0-MeV (kinetic energy) proton enters a 0.20-T field, in a plane perpendicular to the field. What is the radius of its path? (Given: mass of proton= 1.67×10^{-27} kg).

[Ans: 1.8 m]

6. Determine the magnitude and direction of the force between two parallel wires 25 m long and 4.0 cm apart, each carrying 35 A in the same direction.

[Ans: 0.15 N, toward other wire]

7. Determine the magnetic field midway between two long straight wires 2.0 cm apart in terms of the current I in one when the other carries 25 A. Assume these currents are (a) in the same direction, and (b) in opposite directions.

[Ans: (a) $(2.0\times10^{-5}\,\mathrm{T/A})(I - 25\,\mathrm{A})$ (b) $(2.0\times10^{-5}\,\mathrm{T/A})(I + 25\,\mathrm{A})$]

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1) (a) when $I \perp \vec{B}$, estimate $l = 1m$

$$\frac{F_{max}}{l} = IlB$$
$$= 9.4(0.9)$$
$$= 8.46 N/m$$

(b) $\frac{F}{l} = IB\sin\theta$
$$= 9.4(0.9)\sin 35$$
$$= 4.85 N/m$$

4) $F = qvB\sin\theta$, $E = \frac{F}{q}$ (Chapter 1)

$$Eq = qvB\sin\theta$$
$$E = vB\sin\theta$$
$$8.8\times10^{3} = v(7.5\times10^{-3})(\sin 90°)$$
$$v = 1.17\times10^{6} m/s$$

$$r = \frac{mv}{qB} = \frac{9.11\times10^{-31}\cdot1.17\times10^{6}}{1.67\times10^{-19}\cdot7.5\times10^{-3}}$$
$$= 8.4\times10^{-4}$$

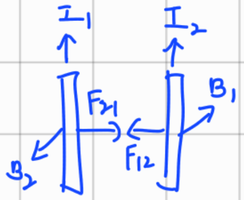
5) $1 MeV = 1.6\times10^{-13} J$, $6.0 MeV = 9.6\times10^{-13} J$

$$k = \frac{1}{2}mv^{2}$$
$$v = \sqrt{\frac{2K}{m}}$$
$$= \sqrt{\frac{2\times9.6\times10^{-13}}{1.67\times10^{-27}}}$$
$$= 3.39\times10^{6}$$

$$r = \frac{mv}{qB}$$
$$= \frac{9.6\times10^{-13}(3.39\times10^{6})}{1.67\times10^{-19}\times0.2}$$
$$=$$

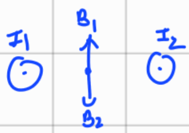
6) $l = 25m$, $d = 0.04$, $r = 0.02m$, $I = 35A$

$$F = l\left(\frac{\mu_0 I_1 I_2}{2\pi d}\right)$$
$$= 25\left(\frac{4\times\pi\times1\times10^{-7}(35)(35)}{2\times\pi\times0.04}\right)$$
$$= 0.153N \text{ attract to each other}$$



7) $d = 0.02$, $I_1 = I$, $I_2 = 25$

(i)



$$\vec{B} = B_1 - B_2$$
$$= \frac{\mu_0(I_1 - I_2)}{2\pi r}$$
$$= \frac{4\times\pi\times1\times10^{-7}}{2\times\pi\times0.01}(I_1 - I_2)$$
$$= \left(2\times10^{-5}(I_1 - 25A)\right)T$$

(ii)

$$\vec{B} = B_1 + B_2$$
$$= \left(2\times10^{-5}(I_1 + 25)\right)T$$

8. A 40.0-cm-long solenoid 1.35 cm in diameter is to produce a field of 0.385 mT at its center. How much current should the solenoid carry if it has 765 turns of wire?

[Ans: 0.160 A]

9. A 2.5-mm-diameter copper wire carries a 33-A current (uniform across its cross section). Determine the magnetic field:
   (a) at the surface of the wire;
   (b) inside the wire, 0.50 mm below the surface;
   (c) outside the wire 2.5 mm from the surface.

[Ans: (a) 5.3 mT (b) 3.2 mT (c) 1.8 mT]

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8) $l = 0.4m$, diameter $= 1.35\times10^{-2}m$, $B = 0.385\times10^{-3}T$, $N = 765$

$$B = \frac{\mu_0 N I}{l} = \frac{4\pi\times10^{-7}(765)(I)}{0.4} = 0.385\times10^{-3}, \quad I = 0.16A$$

9) $d = 2.5\times10^{-3}m$, $R = 1.25\times10^{-3}m$, $I = 33A$

(a) $B = \frac{\mu_0 I}{2\pi R}$

$= \frac{4\pi\times10^{-7}(33)}{2\pi(1.25\times10^{-3})}$

$= 5.28\times10^{-3}T$

(b) $B = \frac{\mu_0 I r}{2\pi R^2}$

$= \frac{4\pi\times10^{-7}(33)(0.75\times10^{-3})}{2\pi(1.25\times10^{-3})^2}$

$= 3.2\times10^{-3}T$

(c) $B = \frac{4\pi\times10^{-7}(33)}{2\pi(3.75\times10^{-3})}$

$= 1.76\times10^{-3}T$

2) $F = I(l\times B)$

$\frac{F}{l} = \vec{I}\cdot\vec{B}$

$\frac{FN}{l\,m} \times \frac{1m}{100cm} = \frac{F}{100l}$

$$\vec{I}\cdot\vec{B} = \begin{vmatrix} i & j & k \\ 3 & 0 & 0 \\ 0.2 & -0.36 & 0.25 \end{vmatrix}$$

$= (0-0)i + (0.75+0)j + (-1.08+0)k$

$= 0.75j - 1.08k$